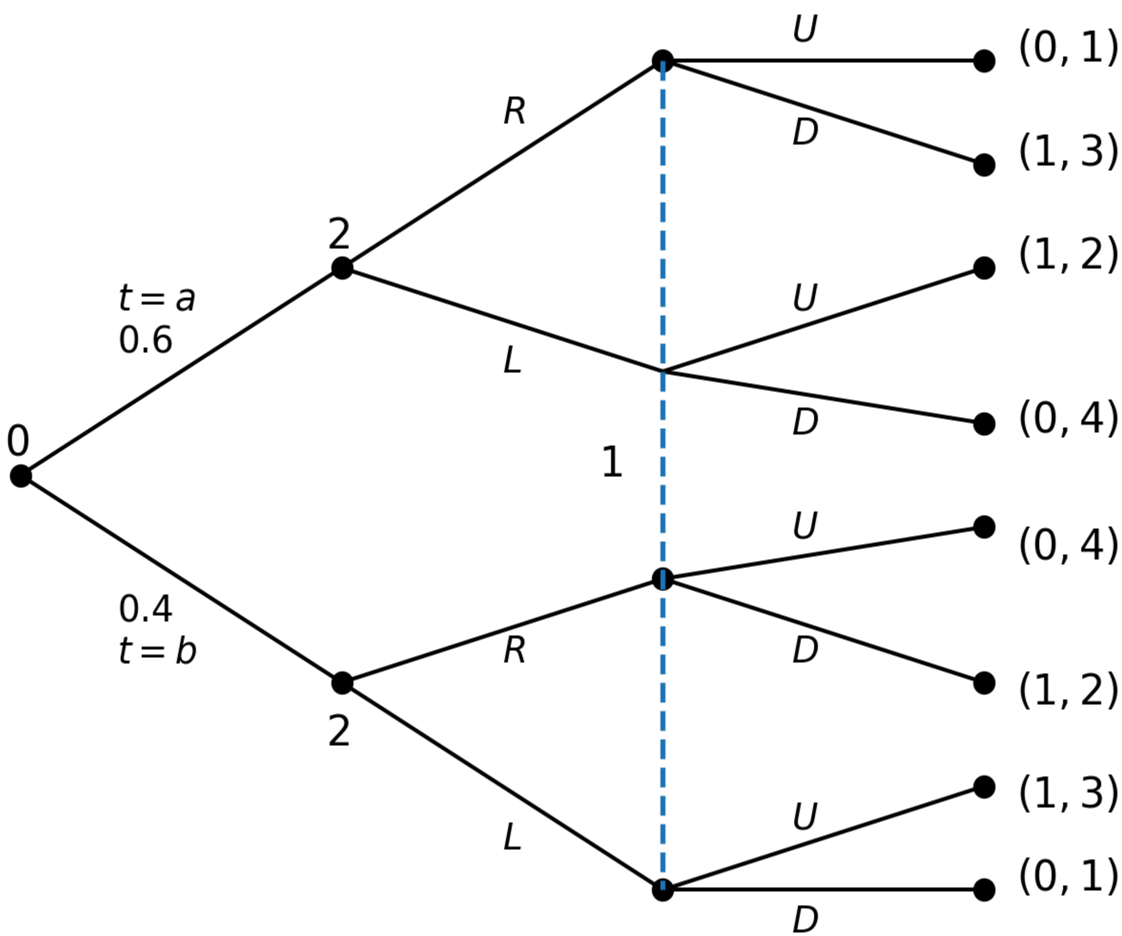
The normal form

Example: A game in extensive form



The normal-form representation

Player 2's strategies are functions. In this case pure strategies can be represented as pairs:

$$[s_2(\{t=a\}) = R, s_2(\{t=b\}) = R] \longrightarrow RR$$

$$[s_2(\{t=a\}) = R, s_2(\{t=b\}) = L] \longrightarrow RL$$

and so on.

	RR	**RL**	**LR**	**LL**
U	$(0, 2.2)$	$(0.4, 1.8)$	$(0.6, 2.8)$	$(1, 2.4)$
D	$(1, 2.6)$	$(0.6, 2.2)$	$(0.4, 3.2)$	$(0, 2.8)$

$$u^1(U, RR) = 0.6 \times 0 + 0.4 \times 0 = 0$$
$$u^2(U, RR) = 0.6 \times 1 + 0.4 \times 4 = 2.2$$
$$u^1(D, RR) = 0.6 \times 1 + 0.4 \times 1 = 1$$
$$u^2(D, RR) = 0.6 \times 3 + 0.4 \times 2 = 2.6$$
$$u^1(U, RL) = 0.6 \times 0 + 0.4 \times 1 = 0.4$$
$$u^2(U, RL) = 0.6 \times 1 + 0.4 \times 3 = 1.8$$
$$u^1(D, RL) = 0.6 \times 1 + 0.4 \times 0 = 0.6$$
$$u^2(D, RL) = 0.6 \times 3 + 0.4 \times 1 = 2.2$$
$$u^1(U, LR) = 0.6 \times 1 + 0.4 \times 0 = 0.6$$
$$u^2(U, LR) = 0.6 \times 2 + 0.4 \times 4 = 2.8$$
$$u^1(D, LR) = 0.6 \times 0 + 0.4 \times 1 = 0.4$$
$$u^2(D, LR) = 0.6 \times 4 + 0.4 \times 2 = 3.2$$

$$
\begin{aligned}
u^1(U, LL) &= 0.6 \times 1 + 0.4 \times 1 = 1 \\
u^2(U, LL) &= 0.6 \times 2 + 0.4 \times 3 = 2.4 \\
u^1(D, LL) &= 0.6 \times 0 + 0.4 \times 0 = 0 \\
u^2(D, LL) &= 0.6 \times 4 + 0.4 \times 1 = 2.8
\end{aligned}
$$

Reduced normal form

- In a normal form game, two of player i's strategies, s_i and s_i', are equivalent if $\forall s_{-i}$, and $\forall j$, $u_j(s_i, s_{-i}) = u_j(s_i', s_{-i})$.

Payoffs for all players must remain unchanged when i uses s_i and switches to s_i'.

- A reduced normal form game is a normal form game in which all equivalent strategies are replaced by a single strategy representing the equivalence set.

Agent-normal form

- Introduced by Selten (1975).
- In an extensive form game it should not matter if every player is replaced by independent agents, one for each of the player's information sets, provided that this agents are given the same final payoffs as the player. At each information set, the agent would make a move.
- Iterative elimination of weakly dominated strategies varies if it is applied to normal form or the agent normal form.

References

Selten, Reinhard. 1975. "Reexamination of the Perfectness Concept for Equilibrium Points in Extensive Games." *International Journal of Game Theory* 4: 25–55.